Noosphere: A Technical Reading List

LLMs, Embedding Geometry, NLP, and the Logic of Coherence

Compiled for Michael Quintin, Theseus

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How to use this document. The list is organized into four technical domains corresponding to the major substrates of Noosphere: (1) the LLM itself, (2) the geometry of the embedding space it produces, (3) the NLP machinery that extracts structured claims from text, and (4) the graph- and logic-based substrate in which coherence is evaluated. A fifth section bridges these to the Quintin Hypothesis directly. Each domain is split into three tiers:

Foundational — assumes essentially no background; builds intuition through visual or verbal explanation, sometimes at the cost of mathematical precision. **Intermediate** — assumes you have read at least one Foundational item in the domain; introduces the formal apparatus. **Advanced / Frontier** — contemporary research, often unsettled; the material Noosphere's design choices live or die by.

Where the resource is freely available, the citation links directly. Paid material is marked **[$]** and is only included where it is genuinely irreplaceable.

A suggested reading path

If you read items in roughly this order, you will not be lost: 3Blue1Brown's *Essence of Linear Algebra* → Alammar's *Illustrated Transformer* → Karpathy's *Let's build GPT from scratch* → Jurafsky & Martin, Chapter 6 (vector semantics) → the Hogan *Knowledge Graphs* survey → Bronstein et al., *Geometric Deep Learning* (skim) → Marks & Tegmark, *The Geometry of Truth*. At that point, you have the conceptual scaffolding to read essentially anything else on this list non-linearly, including the frontier material on interpretability and representation engineering, which is what the Quintin Hypothesis is in direct dialogue with.

1 LLM Internals & Transformer Architecture

What this domain gives you

Noosphere uses Claude (and could use any LLM) for claim extraction, principle distillation, and inference. Without a working model of *how* a transformer produces the embeddings and outputs you are reasoning over, you cannot distinguish a coherence-detection result from an artifact of the model's training distribution. The goal of this section is to make "the LLM" stop being a black box.

Foundational

Grant Sanderson (3Blue1Brown), *Neural Networks* series, YouTube, 2017–2024. https://www.3blue1brown.com/topics/neural-networks

The single best visual introduction to what a neural network is computing. The recent chapters on transformers and attention (2024) are explicitly aimed at viewers with no ML background and are the cleanest available animation of how attention actually moves information between tokens.

Jay Alammar, *The Illustrated Transformer*, blog post, 2018. https://jalammar.github.io/illustrated-transformer/

The canonical first-pass explanation of the original 2017 transformer. Pair it with his follow-ups, *The Illustrated GPT-2* and *The Illustrated Word2vec*. Visual, equation-light, but conceptually faithful.

Stephen Wolfram, *What Is ChatGPT Doing... and Why Does It Work?*, online essay/book, 2023. https://writings.stephenwolfram.com/2023/02/what-is-chatgpt-doing-and-why-does-it-work/

A long, deliberately non-technical exposition that runs from "what is a probability distribution over next tokens" to embeddings and training. Some of Wolfram's framing (e.g., computational irreducibility) is idiosyncratic; treat the philosophical claims as one perspective, not consensus.

Andrej Karpathy, *Let's build GPT: from scratch, in code, spelled out*, YouTube, 2023. https://www.youtube.com/watch?v=kCc8FmEb1nY

You build a small GPT in PyTorch in about two hours. After this video, the phrase "query, key, value" stops being mysterious. His earlier *Zero to Hero* series builds up the prerequisites if needed.

Intermediate

Daniel Jurafsky & James H. Martin, *Speech and Language Processing*, 3rd edition (draft), free online. https://web.stanford.edu/~jurafsky/slp3/

The standard NLP textbook. For LLM internals: Ch. 7 (neural networks), Ch. 9 (RNNs), Ch. 10 (transformers), Ch. 11 (fine-tuning and masked language models), Ch. 12 (model alignment). Always work from the latest dated draft.

Ashish Vaswani et al., *Attention Is All You Need*, NeurIPS 2017. arXiv:1706.03762. https://arxiv.org/abs/1706.03762

The original transformer paper. Now historical, but short and worth reading once with the Alammar illustration open beside it; almost every architecture in production today is a variant.

Lena Voita, *NLP Course | For You*. https://lena-voita.github.io/nlp_course.html

Free, pedagogically excellent course notes covering word embeddings, attention, transfer learning, and analysis/interpretability. Voita's research is also a good entry to model-interpretability papers.

Jared Kaplan et al., *Scaling Laws for Neural Language Models*, arXiv:2001.08361, 2020. https://arxiv.org/abs/2001.08361

Why scale matters. The empirical regularities documented here explain why LLM behaviour is not a bag of tricks but a moderately predictable function of compute, data, and parameters. Read alongside Hoffmann et al. *Training Compute-Optimal Large Language Models* (Chinchilla, arXiv:2203.15556), which corrects Kaplan's data/parameter ratio.

Long Ouyang et al., *Training Language Models to Follow Instructions with Human Feedback* (the InstructGPT paper), arXiv:2203.02155, 2022. https://arxiv.org/abs/2203.02155

RLHF in full. Necessary to understand the gap between a raw next-token predictor and the model Noosphere is actually calling.

Yuntao Bai et al., ***Constitutional AI: Harmlessness from AI Feedback***, arXiv:2212.08073, 2022. https://arxiv.org/abs/2212.08073
Anthropic's alignment approach. Relevant because Noosphere's claim-extraction prompts inherit the constitutional priors of the underlying Claude model — i.e., the LLM is not a neutral observer.

Advanced / Frontier

Nelson Elhage et al., ***A Mathematical Framework for Transformer Circuits***, Anthropic, 2021. https://transformer-circuits.pub/2021/framework/index.html
The conceptual foundation of mechanistic interpretability. Decomposes a transformer into "circuits" of induction heads, attention compositions, etc. Difficult on first read; pairs naturally with Olah et al.'s earlier *Distill* circuits work on vision.

Nelson Elhage et al., ***Toy Models of Superposition***, Anthropic, 2022. https://transformer-circuits.pub/2022/toy_model/index.html
Why a single neuron or dimension can encode multiple features (superposition), and the geometric conditions under which it does. *Directly relevant* to the Quintin Hypothesis: if features are superposed, "the direction for coherence" may be a polysemantic direction, not a clean axis.

Adly Templeton et al., ***Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet***, Anthropic, 2024. https://transformer-circuits.pub/2024/scaling-monosemanticity/
Sparse autoencoders applied to a production model to recover interpretable features. The most concrete demonstration to date that LLMs carry recognizable, human-meaningful directions in their internal representations.

Chris Olah et al., ***Distill: Circuits Thread***, ongoing. https://distill.pub/2020/circuits/
Older (CNN-era) but methodologically foundational; the visual rhetoric of mechanistic interpretability descends from here.

Yonatan Belinkov & James Glass, ***Analysis Methods in Neural Language Processing: A Survey***, TACL 2019. arXiv:1812.08951. https://arxiv.org/abs/1812.08951
Survey of probing, behavioural analysis, and other techniques for asking "what does this model know." Dated, but the taxonomy still maps cleanly onto 2024–2026 work.

2 Embedding Geometry & Vector Spaces

What this domain gives you

This is the mathematical heart of the Quintin Hypothesis. If logical coherence is a geometric property of embeddings, the relevant geometry is high-dimensional, non-Euclidean in places, and almost certainly not flat. You need linear algebra fluency, an understanding of what an "embedding" is formally, and a feel for the modern literature on the linear-representation hypothesis and geometric deep learning.

Foundational

Grant Sanderson, ***Essence of Linear Algebra***, YouTube/3Blue1Brown, 2016. https://www.3blue1brown.com/topics/linear-algebra
Fifteen short videos, geometrically motivated, covering everything from vectors to eigenvectors

and change of basis. Watch this before anything else in this section.

Gilbert Strang, *Linear Algebra and Its Applications* *[$ or library]*, plus his MIT OCW lectures (18.06), free. https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/
The classical introduction. Strang's pedagogical instinct is to keep the geometry visible at all times. The OCW lectures are sufficient even if you don't buy the book.

Jay Alammar, *The Illustrated Word2vec*, blog post, 2019. https://jalammar.github.io/illustrated-word2vec/
Visual introduction to learned distributed representations of words. Even though Noosphere likely uses contextual embeddings (sentence-transformers, OpenAI/ Voyage/Cohere embeddings), the geometric intuitions transfer.

Tomas Mikolov et al., *Efficient Estimation of Word Representations in Vector Space* (word2vec), arXiv:1301.3781, 2013. https://arxiv.org/abs/1301.3781
Historically the result that made everyone start treating embeddings as *semantic* objects. Read alongside the famous follow-up Mikolov et al. 2013b on *linguistic regularities*, which is the origin of the "king − man + woman ≈ queen" phenomenon.

Intermediate

Jurafsky & Martin, *Speech and Language Processing*, Ch. 6: Vector Semantics and Embeddings, free draft. https://web.stanford.edu/~jurafsky/slp3/6.pdf
The definitive textbook treatment: PPMI, SVD, word2vec, GloVe, evaluation. Read this once, even if you've already watched Alammar.

Jeffrey Pennington, Richard Socher, Christopher Manning, *GloVe: Global Vectors for Word Representation*, EMNLP 2014. https://nlp.stanford.edu/pubs/glove.pdf
A different lens on the same problem; valuable because it makes the co-occurrence-matrix-factorization view of embeddings explicit.

Nils Reimers & Iryna Gurevych, *Sentence-BERT: Sentence Embeddings Using Siamese BERT-Networks*, EMNLP 2019. arXiv:1908.10084. https://arxiv.org/abs/1908.10084
How to get a usable *sentence-level* embedding out of a transformer. Almost certainly relevant to Noosphere's ingester, where claims are sentence-to-paragraph in length.

Lloyd N. Trefethen & David Bau, *Numerical Linear Algebra*, SIAM, 1997. *[$]*
The most readable serious treatment of QR, SVD, and Householder reflections (Lecture 10). When your `geometry.py` references Householder reflections, this is where you go to verify what they actually compute and when they are numerically stable. There is no equally good free substitute.

Roger Penrose, *The Road to Reality*, Knopf, 2004, Chapters 11–14. *[$ or library]*
Penrose's chapters on manifolds, fibre bundles, and complex projective spaces are the most readable bridge from undergraduate linear algebra to the differential-geometric language used in geometric deep learning. Skip his later cosmology chapters.

Marco Cuturi & Gabriel Peyré, *Computational Optimal Transport*, Foundations and Trends in ML, 2019. https://arxiv.org/abs/1803.00567
Optional but valuable: optimal transport is the natural language for comparing *distributions* of embeddings (e.g., comparing the embedding-distribution of one person's claims to another's) rather than individual points.

Advanced / Frontier

Michael M. Bronstein, Joan Bruna, Taco Cohen, Petar Veličković, ***Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges***, arXiv:2104.13478, 2021. https://arxiv.org/abs/2104.13478
A 150-page free monograph unifying CNNs, GNNs, and transformers as instances of "learning under symmetry." *The* reference if you take seriously the claim that coherence is a geometric/symmetry property of an embedding manifold. Skim Part I, study Part II, and read Part V (graphs) carefully.

Charles Fefferman, Sanjoy Mitter, Hariharan Narayanan, ***Testing the Manifold Hypothesis***, J. AMS, 2016. arXiv:1310.0425. https://arxiv.org/abs/1310.0425
The mathematically rigorous version of the claim that high-dimensional data tends to live on a low-dimensional manifold. If embeddings live on a manifold of meaning, this paper is where "manifold of meaning" becomes a non-handwaving claim.

Kiho Park, Yo Joong Choe, Victor Veitch, ***The Linear Representation Hypothesis and the Geometry of Large Language Models***, arXiv:2311.03658, 2023. https://arxiv.org/abs/2311.03658
Formalizes the claim that abstract concepts are encoded as linear directions in LLM activations. Treat this as the geometric backbone of Noosphere's hypothesis.

Samuel Marks & Max Tegmark, ***The Geometry of Truth: Emergent Linear Structure in LLM Representations of True/False Datasets***, arXiv:2310.06824, 2023. https://arxiv.org/abs/2310.06824
The single paper most directly aligned with the Quintin Hypothesis: empirically locates linear directions corresponding to factual truth in LLM activation space. Read it with extreme care; replicate it if you can.

Collin Burns, Haotian Ye, Dan Klein, Jacob Steinhardt, ***Discovering Latent Knowledge in Language Models Without Supervision*** (CCS), arXiv:2212.03827, 2022. https://arxiv.org/abs/2212.03827
Contrast-Consistent Search: finds a direction satisfying the logical consistency property $P(x) + P(\neg x) \approx 1$, unsupervised. This is the formal ancestor of any "find the coherence direction" method.

Andy Zou et al., ***Representation Engineering: A Top-Down Approach to AI Transparency***, arXiv:2310.01405, 2023. https://arxiv.org/abs/2310.01405
Practical recipes for finding and intervening on concept directions inside an LLM. The applied counterpart to Park/Marks/Tegmark.

Tatsunori Hashimoto, David Alvarez-Melis, Tommi Jaakkola, ***Word Embeddings as Metric Recovery in Semantic Spaces***, TACL 2016. https://aclanthology.org/Q16-1019/
Treats word embeddings as solutions to a metric-recovery problem. Useful if you ever try to put Noosphere's coherence metric on a principled geometric footing rather than an empirical one.

3 NLP Information Extraction

What this domain gives you

`ingester.py` converts unstructured transcript text into typed claims, relationships, and (in principle) principles. The quality of every downstream coherence judgement is bounded above by the quality of this step. This section covers the parsing and extraction stack that lives between raw text and the knowledge graph.

Foundational

Ines Montani & Matthew Honnibal, *Advanced NLP with spaCy*, free online course. https://course.spacy.io/
Written by the people who built spaCy. Covers tokenization, POS tagging, dependency parsing, NER, rule-based matching, and training custom models. If `ingester.py` has a spaCy fallback, this is where you understand what that fallback is actually doing.

Steven Bird, Ewan Klein, Edward Loper, *Natural Language Processing with Python* (the NLTK book), free online. https://www.nltk.org/book/
Dated (it predates the transformer era) but it remains the friendliest introduction to the classical NLP pipeline: tokenization, chunking, tagging, parsing, relation extraction. Read Ch. 7 (information extraction) specifically.

Jurafsky & Martin, Ch. 18 (Information Extraction), free. https://web.stanford.edu/~jurafsky/slp3/18.pdf
The current canonical treatment: NER, relation extraction, event extraction, template filling, temporal expression handling. Read alongside Ch. 19 (Question Answering) and Ch. 20 (Coreference).

Intermediate

Jacob Devlin, Ming-Wei Chang, Kenton Lee, Kristina Toutanova, *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*, NAACL 2019. arXiv:1810.04805. https://arxiv.org/abs/1810.04805
Why a single pretrained encoder underlies essentially every modern extraction system, including the NER and relation-extraction components you might bolt on to spaCy.

Kenton Lee et al., *End-to-end Neural Coreference Resolution*, EMNLP 2017. arXiv:1707.07045. https://arxiv.org/abs/1707.07045
Coreference (deciding when "she," "the founder," and "Michael" refer to the same entity) is foundational to building a knowledge graph that isn't a mess of duplicate nodes.

Pere-Lluís Huguet Cabot & Roberto Navigli, *REBEL: Relation Extraction By End-to-end Language generation*, EMNLP Findings 2021. https://aclanthology.org/2021.findings-emnlp.204/
A transformer-based relation extractor that frames extraction as generation. Practical and reasonably easy to deploy.

Christina Niklaus, Matthias Cetto, André Freitas, Siegfried Handschuh, *A Survey on Open Information Extraction*, COLING 2018. https://aclanthology.org/C18-1326/
Open IE is the line of work that produces tuples like `(Subject, Relation, Object)` without a fixed schema. Directly relevant to building a knowledge graph whose ontology is not known a priori.

Advanced / Frontier

Somin Wadhwa, Silvio Amir, Byron C. Wallace, *Revisiting Relation Extraction in the era of Large Language Models*, ACL 2023. arXiv:2305.05003. https://arxiv.org/abs/2305.05003

How well do LLMs do information extraction zero- and few-shot, and where do they fail? Required reading if Noosphere is asking Claude to extract relations directly.

Shuofei Qiao et al., *Reasoning with Language Model Prompting: A Survey*, ACL 2023. arXiv:2212.09597. https://arxiv.org/abs/2212.09597

Survey of prompting techniques (chain-of-thought, self-consistency, decomposition, tool use). Most "extract these claims from this transcript" prompts are crude versions of techniques surveyed here.

Patrick Lewis et al., *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*, NeurIPS 2020. arXiv:2005.11401. https://arxiv.org/abs/2005.11401

The original RAG paper. Relevant to Noosphere because reasoning over the knowledge graph is, in effect, retrieval-augmented inference.

Christian Stab & Iryna Gurevych, *Parsing Argumentation Structures in Persuasive Essays*, Computational Linguistics 2017. https://aclanthology.org/J17-3005/

Argument mining — identifying claims, premises, and their support relations in text — is the closest extant subfield to what Noosphere wants to do at the principle level. Their later book is paywalled but this paper is freely available and surprisingly readable.

Manuel Lardelli et al., recent surveys on *LLM-based Knowledge Graph Construction* (multiple papers, 2023–2025).

Search Google Scholar for "LLM knowledge graph construction survey 2024." This is an actively moving target; pin a few recent surveys, don't try to read everything.

4 Knowledge Graphs, Logic, and Coherence

What this domain gives you

The ontology layer (`ontology.py`) is a graph. The coherence engine (`coherence.py`) is, in effect, applying logical and probabilistic operations over that graph. This section covers (a) graph theory and knowledge graphs, (b) classical and non-classical logic — including paraconsistent logic, which you need if your system is to encounter and not collapse under contradictions, and (c) coherentism as a position in epistemology, with its computational implementations.

Foundational

Aidan Hogan et al., *Knowledge Graphs*, ACM Computing Surveys 2021 / Synthesis Lectures 2022. arXiv:2003.02320. https://arxiv.org/abs/2003.02320

The reference survey. Covers data models, schemas, identity, context, deductive and inductive knowledge, quality, and refinement. Long; treat as a handbook.

NetworkX documentation and tutorials, free. https://networkx.org/documentation/stable/tutorial.html

`ontology.py` uses NetworkX. Spend an afternoon with the tutorial and a few of the algorithm reference pages (especially shortest path, centrality, connected components, and graph traversal). You will write better graph code after this than after months of guessing.

Reinhard Diestel, *Graph Theory*, 5th ed., free PDF on the author's site. https://diestel-graph-theory.com/

The standard graduate graph theory text. Free download. Don't read it linearly; treat as reference for any graph-theoretic concept Noosphere needs (cycles, bipartiteness, connectivity, planarity).

Stanford Encyclopedia of Philosophy, *Coherentist Theories of Epistemic Justification*, free online. https://plato.stanford.edu/entries/justep-coherence/
The accessible entry point to the philosophical literature you are operationalizing. Read the companion entries on *Foundationalist* justification and *Truth* alongside it.

Stanford Encyclopedia of Philosophy, *The Coherence Theory of Truth*, free. https://plato.stanford.edu/entries/truth-coherence/
A briefer entry on the related but distinct claim that truth itself just is coherence among beliefs. Noosphere's hypothesis lives in the space between coherence-as-justification and coherence-as-truth; you should know exactly where you sit.

Intermediate

Antoine Bordes et al., *Translating Embeddings for Modeling Multi-relational Data* (TransE), NeurIPS 2013. https://proceedings.neurips.cc/paper/2013/hash/1cecc7a77928ca8133fa24680a88d2f9-Abstract.html
The paper that started the modern field of knowledge-graph embeddings, in which entities and relations live in the same vector space. Foundational for any attempt to unify Noosphere's graph and embedding views.

Quan Wang, Zhendong Mao, Bin Wang, Li Guo, *Knowledge Graph Embedding: A Survey of Approaches and Applications*, IEEE TKDE 2017. https://ieeexplore.ieee.org/document/8047276 (preprint also widely circulated).
The standard survey of KG-embedding methods up through 2017 — TransE, TransR, DistMult, ComplEx, RotatE descendants. Foundational vocabulary for the 2020s literature.

Franz Baader, Diego Calvanese, Deborah McGuinness, Daniele Nardi, Peter Patel-Schneider (eds.), *The Description Logic Handbook*, 2nd ed., Cambridge, 2007. *[$ or library]*
Description logic is the formal underpinning of OWL and most serious ontology languages. Chapters 1–3 give you the vocabulary (TBox, ABox, subsumption, satisfiability) you will repeatedly need if you formalize Noosphere's ontology.

Graham Priest, *An Introduction to Non-Classical Logic: From If to Is*, 2nd ed., Cambridge, 2008. *[$ or library]*
The single best one-volume entry point to modal, intuitionistic, many-valued, relevant, and paraconsistent logics. Noosphere will encounter contradictions; you need to decide *which* non-classical logic governs how it reasons about them rather than defaulting to classical logic and accepting explosion.

Graham Priest, Koji Tanaka, Zach Weber, *Paraconsistent Logic*, Stanford Encyclopedia of Philosophy, free. https://plato.stanford.edu/entries/logic-paraconsistent/
The short version of why "a contradiction implies everything" is a feature of classical logic specifically, and how paraconsistent systems avoid it.

Paul Thagard, *Coherence in Thought and Action*, MIT Press, 2000. *[$ or library]*
The most thoroughly worked-out computational theory of coherence in the literature. Thagard formalizes coherence as a constraint-satisfaction problem over a network of elements and constraints, solvable by connectionist or algorithmic methods. The conceptual ancestor of any coherence *engine* worth building.

Advanced / Frontier

Luc Bovens & Stephan Hartmann, *Bayesian Epistemology*, Oxford, 2003. *[$ or library]*
A rigorous probabilistic treatment of coherence. They prove the (surprising) result that "more coherent" does not in general entail "more probable," and discuss the conditions under which it does. Read *before* claiming that Noosphere's coherence scores track probability of truth.

Erik J. Olsson, *Against Coherence: Truth, Probability, and Justification*, Oxford, 2005. *[$ or library]*
The most influential critical examination of coherentism in recent decades. You should be able to articulate Olsson's negative results before claiming Noosphere's methodology is sound.

Laurence BonJour, *The Structure of Empirical Knowledge*, Harvard, 1985. *[$ or library]*
The classical defence of coherentism in 20th-century epistemology. BonJour later retracted parts of his position; that retraction is itself instructive and is discussed in his 1999 paper *The Dialectic of Foundationalism and Coherentism*.

Achille Varzi, *Mereology*, Stanford Encyclopedia of Philosophy, free. https://plato.stanford.edu/entries/mereology/
The formal theory of parts and wholes. Underlies any serious treatment of "principle distillation" — when is one claim a part of another, when is a principle a whole composed of claims?

Sven Ove Hansson, *Logic of Belief Revision*, Stanford Encyclopedia of Philosophy, free. https://plato.stanford.edu/entries/logic-belief-revision/
The AGM framework for how a rational agent's beliefs should change in response to new information. The natural formal home for Noosphere's `temporal.py` module.

Aditya Pal et al. / various authors, *Knowledge Graphs and Large Language Models: A Survey of Unified Approaches*, multiple 2023–2025 surveys.
Search "unifying KGs and LLMs survey" on arXiv. The field is moving; pin two or three recent surveys rather than committing to one. Pan, Luo et al. *Unifying Large Language Models and Knowledge Graphs: A Roadmap* (arXiv:2306.08302, 2023) is a good entry point.

Maximilian Nickel et al., *A Review of Relational Machine Learning for Knowledge Graphs*, Proc. IEEE 2016. arXiv:1503.00759. https://arxiv.org/abs/1503.00759
Older but still the cleanest synthesis of relational learning, latent-feature models, and graph-feature models. Necessary if Noosphere ever does inductive inference over the graph itself rather than over the embedding alone.

5 Bridging Section: Reading Directly Aligned with the Quintin Hypothesis

These items are not in a separate domain; they are the ones that, taken together, define the research frontier the Quintin Hypothesis is in direct dialogue with. If you read nothing else from this list, read these.

Direct alignment with the hypothesis

Marks & Tegmark, *The Geometry of Truth* (already cited).
Empirical claim: factual truth has a roughly linear direction in LLM activation space.

Burns, Ye, Klein, Steinhardt, *Discovering Latent Knowledge in Language Models Without Supervision* (CCS) (already cited).
Empirical claim: the direction satisfying logical-consistency constraints $P(x) + P(\neg x) \approx 1$

can be found without labels. Operationally, *the closest thing in the literature to a "coherence detector"*.

Park, Choe, Veitch, *The Linear Representation Hypothesis* (already cited).
Provides the theoretical underpinning for treating concepts (including "coherence," if it is a concept) as linear directions.

Elhage et al., *Toy Models of Superposition* (already cited).
The most important *cautionary* paper. If features are superposed, then "the direction for coherence" may be a polysemantic projection of several things at once, and naive measurements will conflate them.

Zou et al., *Representation Engineering* (already cited).
Toolkit for actually performing the kind of direction-finding, direction-steering, and direction-evaluation Noosphere wants to do.

Stephanie Lin, Jacob Hilton, Owain Evans, *TruthfulQA: Measuring How Models Mimic Human Falsehoods*, ACL 2022. arXiv:2109.07958. https://arxiv.org/abs/2109.07958
The benchmark dataset most commonly used to evaluate "truth" in LLMs. Whatever Noosphere claims about truth needs to be checkable against something; this is the closest existing standard.

Kenneth Li et al., *Inference-Time Intervention: Eliciting Truthful Answers from a Language Model*, NeurIPS 2023. arXiv:2306.03341. https://arxiv.org/abs/2306.03341
Finds and intervenes on attention-head directions correlated with truthfulness. The applied counterpart to CCS.

Paul Thagard, *Coherence as Constraint Satisfaction*, Cognitive Science 1998 (and the 2000 MIT Press book of the same theme). https://onlinelibrary.wiley.com/doi/10.1207/s15516709cog2201_1
The clearest statement of the computational coherentist program. If you have not formally compared Noosphere's six-layer scoring engine to Thagard's constraint-satisfaction formulation, you should.

Catherine Olsson, Nelson Elhage et al., *In-context Learning and Induction Heads*, Anthropic 2022. https://transformer-circuits.pub/2022/in-context-learning-and-induction-heads/index.html
A specific, well-documented circuit-level finding about how LLMs do certain kinds of pattern completion. Worth reading because it is one of the few cases where a behavioural capability has been traced to identifiable geometric/algorithmic structure inside the model.

6 Appendix: Prerequisite Math Primers

If any of the following feels unfamiliar, fill it in *before* attempting the Advanced tier of any domain. Each item is freely available.

Probability and statistics. Joseph Blitzstein & Jessica Hwang, *Introduction to Probability*, 2nd ed., free draft and full Harvard Stat 110 lectures. https://projects.iq.harvard.edu/stat110

Multivariable calculus. MIT OCW 18.02, free. https://ocw.mit.edu/courses/18-02-multivariable-calcul

Information theory. David MacKay, *Information Theory, Inference, and Learning Algorithms*, Cambridge, 2003, free PDF. https://www.inference.org.uk/itila/book.html
Particularly chapters 1–6 (information theory) and chapters on inference. The single most useful free book on the list if you want to formalize what "information," "surprise," and "coherence" mean

quantitatively.

Optimization. Stephen Boyd & Lieven Vandenberghe, *Convex Optimization*, free PDF. https://web.stanford.edu/~boyd/cvxbook/
Essentially everything trained in ML reduces to some optimization problem; Boyd is the canonical reference.

A note on currency

The interpretability and representation-engineering literature in particular moves fast — six-to-twelve-month half-lives are common. Anchor yourself on the canonical references above, but make a habit of skimming `transformer-circuits.pub` (Anthropic), `distill.pub` (Olah and collaborators), and the `arXiv` categories `cs.CL`, `cs.LG`, and `cs.AI` weekly. For knowledge-graph × LLM work, the `ACL Anthology` and `ISWC` proceedings are the authoritative venues.